## GO / NO-GO GATES

### Gate 1 — mixing & continuity → NO-GO

Unintegrated iLISI 1.0 of 7 (complete per-patient segregation); Leiden clusters 99.3% patient-pure; patient identity 99.5% kNN-predictable. Mixing appears only when Harmony is forced (θ≥3), collapsing patient centroids rather than revealing a shared axis. Per-patient archipelago — no continuous, sample-mixed malignant manifold exists to fit.

### Gate 2 — module geography → NA

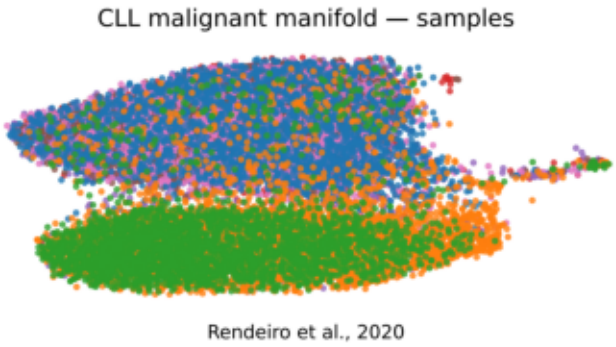
Not evaluated — gate chain stops at the Gate 1 NO-GO (§6).

### Gate 3 — hub cross-repression → NA

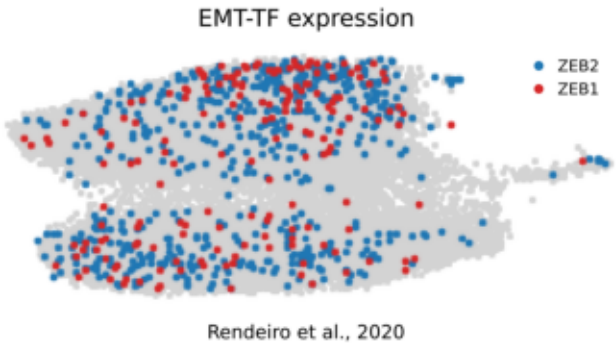
Not evaluated — branching-only; no trajectory fit.

## STEP FIGURES

**Manifold — samples**




**EMT-TF expression**




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**Dataset.**
Chronic Lymphocytic Leukemia (Rendeiro et al., 2020), 10x — 18,657 malignant cells · 7 samples / 4 patients.

**Verdict.**
Pattern does NOT reproduce. The malignant compartment is a set of independent per-patient CLL clones, not a continuous developmental manifold. NO-GO at Gate 1; no trajectory, no bifurcation. Descriptive EMT-TF overlay: ZEB2 (12.4%), ZEB1 (3.8%).

## Parameter choices

| | | | |
|---|---|---|---|
| **Sample threshold** | ≥100 cells → 7 of 11 samples (18,657 cells); largest multiplicative count gap 1012→80 (12.6×) | **n_pcs** | 20 (scree elbow at 6, floored to atlas robust range) |
| **Integration** | Harmony on sample, swept 0→5; protocol-default θ=2 shown; higher over-integrates | **Island removal** | none (disconnected components are whole patients, not noise) |
| **Geometry** | NA (no trajectory); manifold not sample-mixed; Gate 1 NO-GO | **EMT-TFs** | ZEB2 (12.4%), ZEB1 (3.8%); SNAI2/TWIST/FOXC2/PRRX2 absent |